WebClient Computing

Computer Science Department
Kyungyong Lee - 이경용

Instructor Introduction

- Kyungyong Lee - 이경용
  - Assistant professor in KMU Computer Science since 2016 March.
  - Graduated with PH.D. in the University of Florida in 2014
  - Thesis: Large-scale distributed computing resource management
  - Few industry experiences as a software development engineer / Researcher
    - Amazon Web Services
    - Hewlett Packard Labs.
    - Samsung

To Better Understand You

- Knows HTML document is organized
- Knows how Javascript works
- Programming experiences
  - Java, C, Python, …
  - Preferred dev. Environment: IDE, vim, …
  - Operating system
- Owns own homepage
- Knows how HTTP works
- Heard of TCP/IP
- Can explain difference between TCP and UDP

About This Course

- To provide high-level overview of how web-based system works
- Helps you to understand web markup language
  - HTML, Javascript, …
- Build a dynamic homepage on your own
- **Have better understanding of web and the Internet**

You Are Expected to Make

- Not a product-ready web page
- A page that contains core techniques to build a professional web page

Course Logistics

- 7호관 2층 32호실 - Thur. 16:30 ~ 17:45
- 7호관 4층 47호실 - Fri. 10:30 ~ 11:45
- Recommend attendance and class participation
- Please NO cell phone use during the class
- English is the primary language of this course
- Students are encouraged to talk/ask in Korean – Don’t hesitate to ask questions!!!

Logistics : Communication

- Notices and Q&A using slack
  - https://kmu-cs-web-client-161.slack.com
  - Need to create an account with **@kookmin.ac.kr** unless you already have one
  - Iphone and andrioid - search for “slack”
- E-mail
  - leeky@kookmin.ac.kr
- Cell phone: 010-6539-0619
- Office hour
  - Location. 생활관 B동 614호
  - Tues. 13:30 ~ 14:30
  - Wed. 14:00 ~ 15:00
  - After the course when the instructor is available
  - By appointment

Grades (Evaluation)

- Midterm: 30%
- Final: 30%
- Project: 20%
- Homework: 15%
- Attendance: 5%
  - Per KMU policy, missing classes more than or equal to **seven** times will fail

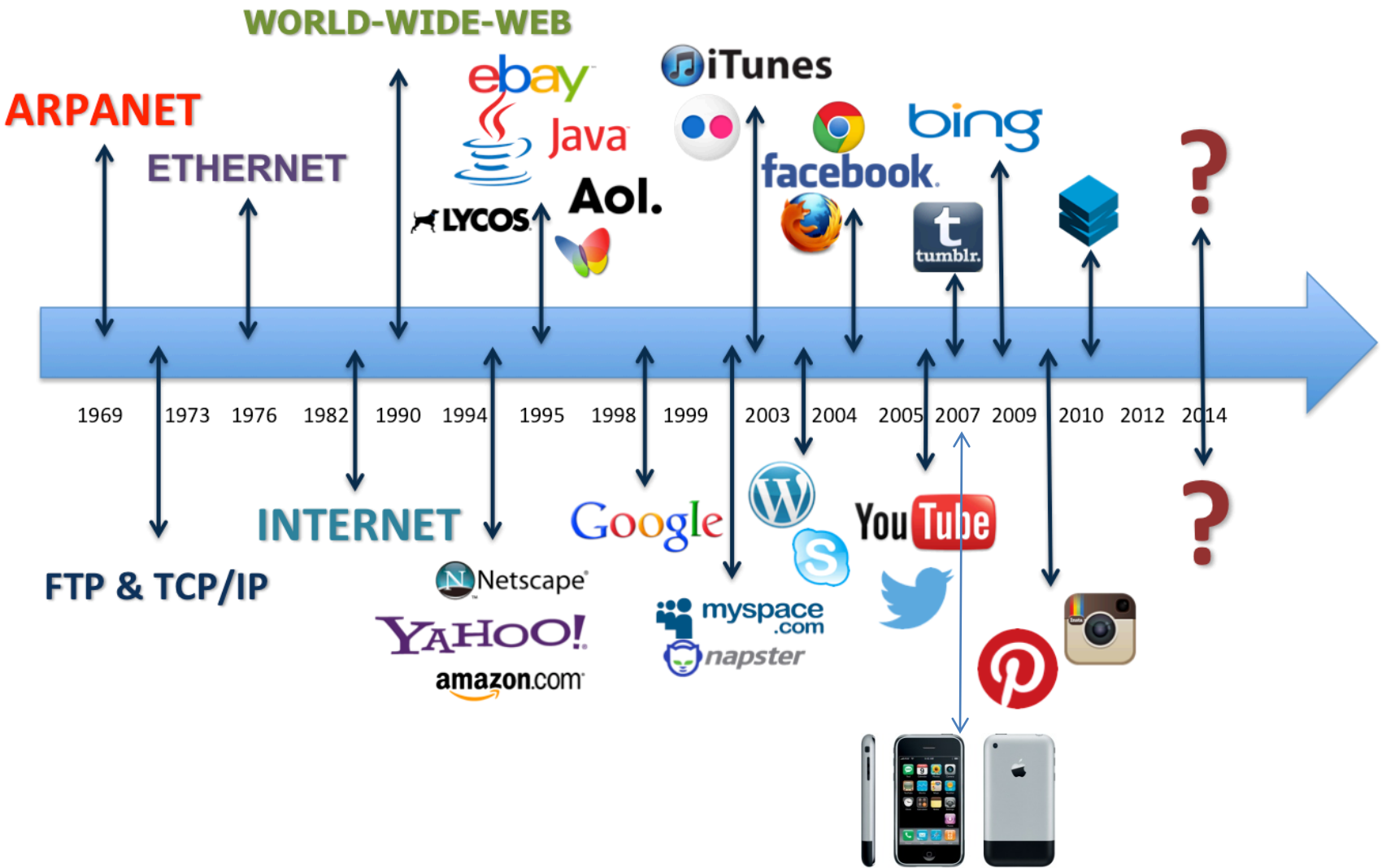
Project & Assignments

- A course project to create your own client program using techniques learned from the course
- 2~3 programming and written homework
- Let's respect others' hard work!!!

The History of the Internet

- Connected mainframes, devices that use internet protocol suites
  - Defines how to communicate among devices
  - TCP/IP is widely in use
- The Internet is a huge connection of devices over the communication network
- The first Internet
  - ARPANET from UCLA in 1969
  - Packet switching network
  - UCLA, UCSB, Stanford, Univ. of Utah

Events of the Internet



Network Protocols

- A set of rules to decide how different entities talk to each other
- Internet Protocol (IP)
  - Identifies the address of each entity
  - Sequence, reliability is not guaranteed
- Transmission Control Protocol (TCP)
  - Delivering messages to the appropriate program in a computer
  - Guaranteed sequence and reliability
- Hyper Text Transport Protocol (HTTP)
  - Decide how to organize and parse contents
- Details will be covered in other courses
  - Computer networks, communications

IP Address

- Identifies a computer in a network
  - Similar to a mail address
  - Numeric form: 14.101.49.15
  - IPV4: 32 bit
- IPV4 address runs out and IPV6 is widening its application
  - IPV6: 128 bit

TCP/IP Comparing to Mail

- IP layer is similar to mail delivery person
  - The delivery person does not know if the mail is lost
  - The delivery person does not know what the content is
  - The delivery person does not take who the recipient is (mother, father, ??)
- TCP layer assures correct delivery
  - Sender and receiver can make sure the delivery succeeds (by calling each other)
  - In case of failure, resend it.
  - If the sending amount is too much, control it
  - Deliver to the correct person
- **With TCP, application can assume the message is correct**

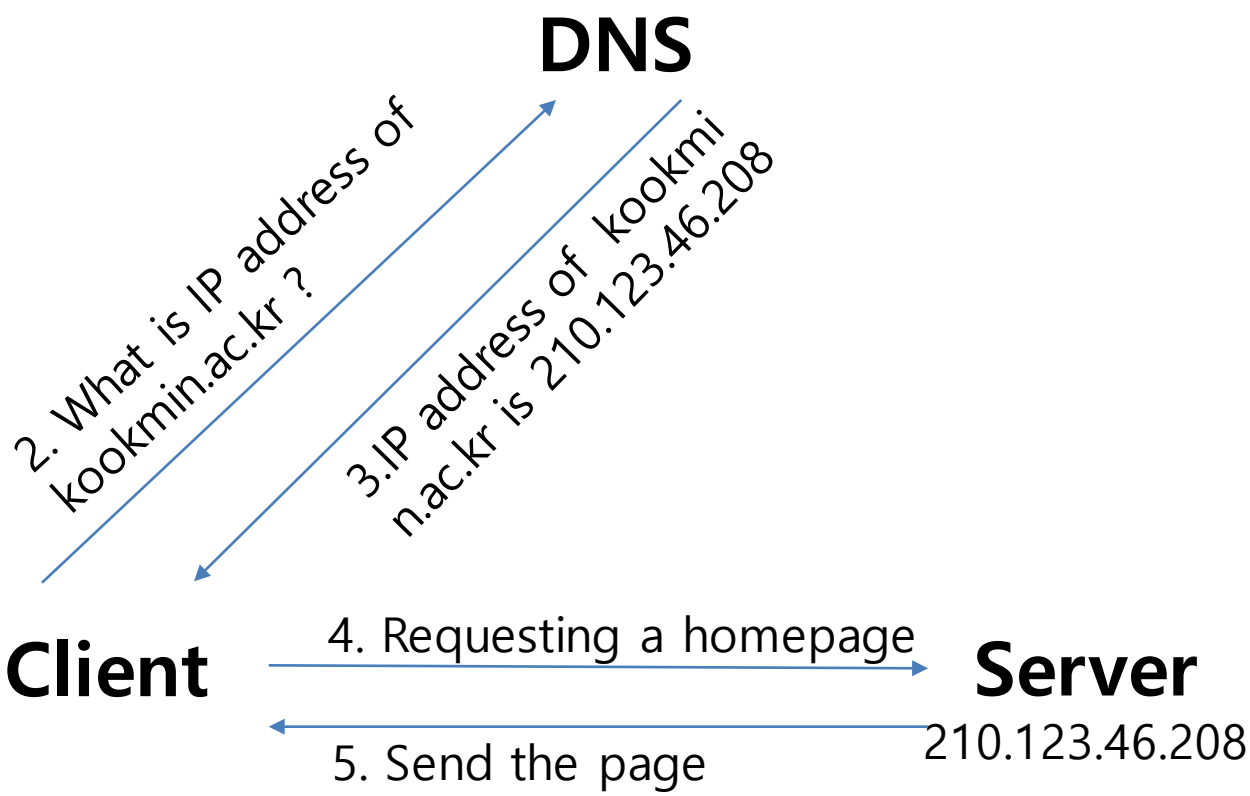
Domain Name Service (DNS)

- Helps people to better identify/access server's address
  - kookmin.ac.kr : 210.123.46.208
- DNS is responsible for translating the numeric IP address to textual address

Communication Model

- Server-client model
  - Server: Content provider
    - Naver, Daum, Google, Facebook
  - Client: Content consumer
    - Internet explorer, cell-phone browser
  - Classical approach and straightforward
  - Shortcomings
    - Scalability
    - Single-point-of-failure
- P2P model
  - Each device is a server and client
  - BitTorrent
  - Quality of Service uncertainty

Summary



1. Types kookmin.ac.kr in a browser

Step 4 and 5 can be repeated in case of packet lost (TCP)

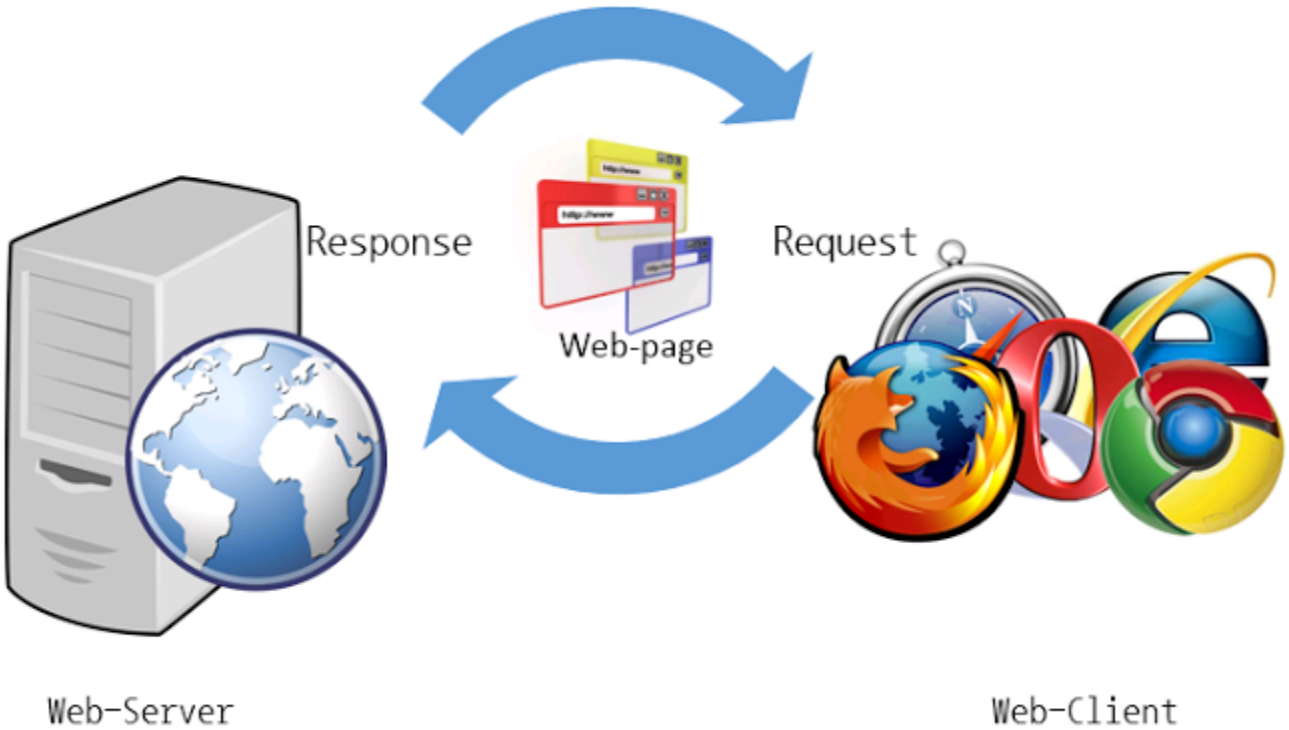
World Wide Web

- An open information space identified by the URL
- Interlinked by hypertext links operated by a click
- Can be accessed via the Internet

Composition of the Web

- Information is stored in a form of **web page**
  - Using HTML
- The webpage is stored and served by **web server** using Apache server or Internet Information Server (IIS)
- **Web client** access the web page using a web browser
  - Internet explorer or chrome, …

How Web Page is served



KMU CS
WebServer Computing

KMU CS
WebClient Computing

Summary So Far…

- The Internet
  - A set of devices inter-connected
  - Defines rules to communicate among them
  - Ethernet, TCP/IP, HTTP, …
  - Details in Computer Network course
- The Web
  - A collection of information connected via hyperlinks
  - Pages sent/received over the Internet
  - Composed of Server and Client
    - Details in WebServer and **WebClient** Computing

Web Client and Web Browser

- Web browser is a software application that **retrieves**, **presents**, **traverses** web pages in the World Wide Web

Usage share of all browsers for October – December 2015

Source	Chrome + Android	Internet Explorer	Firefox	Safari	Opera	Others
US government	42%	24.2%	9.6%	19.8%	††	4.3%
StatCounter	52.76%	10.27%	9.3%	12.39%	5.66%††	2.89%

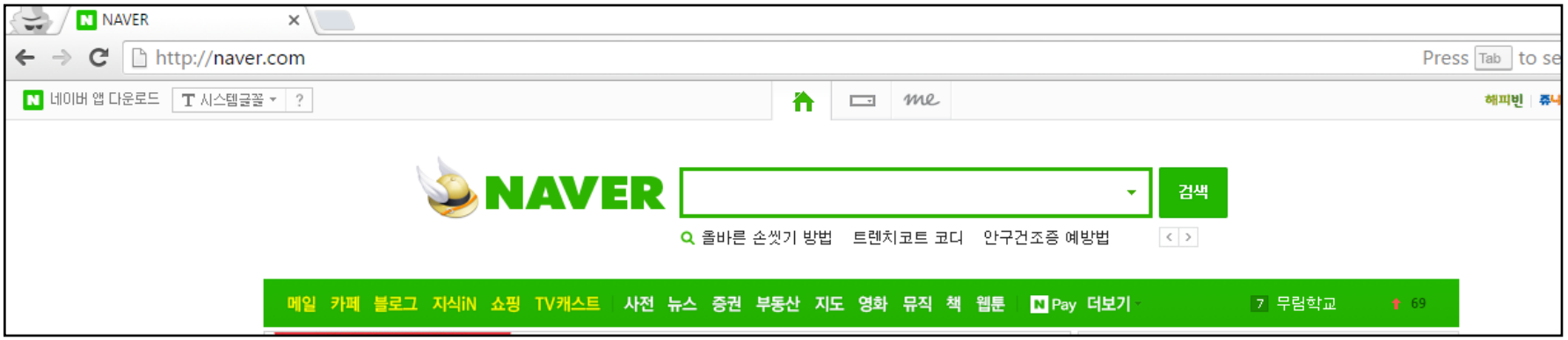
Usage share of desktop browsers for October – December 2015

Source	Chrome	Internet Explorer	Firefox	Safari	Opera	Others
NetMarketShare	31.3%	50.0%	11.9%	4.6%	1.5%	0.6%

Usage share of web browsers – from wikipeida

How Web Browser Talks to Servers

- In a web browser, we type an address



- Destination address is expressed using URL

URL

- Uniform Resource Locator
- A reference to web source on the network
- The format
  - scheme:[//[user:password@]host[:port]][/]path[?query][#fragment]
    - Scheme – Express a protocol how the request/response is formed
    - Host – Address for the server
    - Path – The location where the target page is located in the server

URL Example

- http://naver.com/news
  - Scheme: http
  - Host: naver.com
  - Path: news
- More about the scheme
  - Various protocols for different purposes
  - http, ftp, mailto, …

HTTP Scheme

- Hyper Text Transport Protocol
- A protocol to guide communication between client and server
- Hyper text is a structured text that uses links to navigate through different pages
  - Generally refers to HTML files
- HTTP uses port 80
  - Port number can be omitted in the URL

HTTP Example

- Type - http://naver.com/news and it translates into the following command

Request

```
GET /news HTTP/1.1
Host: www.naver.com
Connection: keep-alive
Accept: text/html,application/xh
Upgrade-Insecure-Requests: 1
User-Agent: Mozilla/5.0 (Windows
Accept-Encoding: gzip, deflate,
Accept-Language: en-US,en;q=0.8
```

Response

```
HTTP/1.1 200 OK
Server: nginx
Date: Mon, 29 Feb 2016 14:22:49 GMT
Content-Type: text/html; charset=UTF-8
Transfer-Encoding: chunked
Connection: close
Cache-Control: no-cache, no-store, must
Pragma: no-cache
P3P: CP="CAO DSP CURa ADMa TAIa PSAa OU
Content-Encoding: gzip
X-Frame-Options: SAMEORIGIN
```

Web Page Structure

- HTML
  - Content of a page and its organization
- CSS
  - Appearance (How it looks)
    - Background color, cell spacing, font size, …
- Javascript
  - Functions for software programming feature
- WebClient renders a page using the above

HTML

- Hyper Text Markup Language
- From HTML slide
  - Hyper text is a structured text that uses links to navigate through different pages
- Markup language
  - Represents structure of documents or data
  - Different from programming language that executes with a computer (Java)
- Express title, sentence, list, link, references, …

Example of HTML

```
<!DOCTYPE html>
<html>
<head>
    <title>My first web page</title>
    <!-- This is a comment, by the way -->
</head>
<body>
<h1>My first web page</h1>
<h2>What this is</h2>
<p>A simple page put together using HTML. <em>I said a simple page put together using HTML.</em> A simple page put together using
page put together using HTML. A simple page put together using HTML. A simple page put together using HTML. A simple page put tog
HTML.</p>
<h2>Why this is</h2>
<ul>
    <li>To learn HTML</li>
    <li>
        To show off
        <ol>
            <li>To my boss</li>
            <li>To my friends</li>
            <li>To my cat</li>
            <li>To the little talking duck in my brain</li>
        </ol>
    </li>
    <li>Because I have fallen in love with my computer and want to give her some HTML loving.</li>
</ul>
<h2>Where to find the tutorial</h2>
<p><a href="http://www.htmldog.com"><img src="http://www.htmldog.com/badge1.gif" width="120" height="90" alt="HTML Dog"></a></p>
<h3>Some random table</h3>
<table>
    <tr>
        <td>Row 1, cell 1</td>
        <td>Row 1, cell 2</td>
        <td>Row 1, cell 3</td>
    </tr>
```

Reference

- http://malonemediagroup.com/history-of-the-internet-timeline-an-ever-evolving-digital-world/
- http://wiznetacademy.com/?p=2886
- Web Application Software Development by C.C. Cheung